COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 212 - Automating Server Management	**SECTION:**	CPE31S2
DATE PERFORMED:	08/13/2026	**DATE SUBMITTED:**	08/13/2026
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Prelim Exam

1. PROCEDURE OUTPUT

SETUP

Hostname	Operating System	Ip Address
Ubuntu1	Ubuntu	192.168.56.120
Ubuntu2	Ubuntu	192.168.56.121
centos1	Cent OS	192.168.56.122

Part A: Installing and Verifying Ansible

1. Update the package repository.
2. Install pip if it is not yet installed.
3. Install Ansible.
4. Verify that Ansible is installed correctly.
5. Verify SSH connectivity to each managed node.

Output:

Ansible Version

```
alvarezworkstation@workstation:~$ ansible --version
ansible 2.10.8
  config file = None
  configured module search path = ['/home/alvarezworkstation/.ansible/plug
dules', '/usr/share/ansible/plugins/modules']
  ansible python module location = /usr/lib/python3/dist-packages/ansible
  executable location = /usr/bin/ansible
  python version = 3.10.12 (main, Jun 22 2026, 18:55:27) [GCC 11.4.0]
alvarezworkstation@workstation:~$
```

SSH connectivity from control node to ubuntu 1:

```
alvarezworkstation@workstation:~$ ssh alvarezworkstation@192.168.56.1
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 6.8.0-134-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

2 additional security updates can be applied with ESM Apps.
Learn more about enabling ESM Apps service at https://ubuntu.com/esm

Last login: Thu Jul 30 08:13:58 2026 from 192.168.56.119
alvarezworkstation@server1:~$
```

SSH connectivity from control node to ubuntu2:

```
alvarezworkstation@workstation:~$ ssh alvarezworkstation@192.168.56.
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 6.8.0-134-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

2 additional security updates can be applied with ESM Apps.
Learn more about enabling ESM Apps service at https://ubuntu.com/esm

Last login: Thu Jul 30 08:14:53 2026 from 192.168.56.119
alvarezworkstation@server2:~$
```

Part B: Building an Inventory File

Inventories provide Ansible with information about managed nodes and their network locations.

1. Create a working directory
2. Create an inventory file
3. Add the following content.
4. Verify the inventory structure.
5. Ping all managed nodes.

Demonstrate the following:

- inventory.yaml

alvarezworkstation@ubuntu1: ~/ansible-lab

GNU nano 6.2 inventory.yml

```
all:
  children:
    ubuntu:
      hosts:
        ubuntu1:
          ansible_host: 192.168.56.120
        ubuntu2:
          ansible_host: 192.168.56.121
    centos:
      hosts:
        centos1:
          ansible_host: 192.168.56.122
          ansible_user: ealvarez
```

- Successful inventory verification

```
alvarezworkstation@ubuntu1:~/ansible-lab$ ansible-inventory -i inventory.yml --list
{
    "_meta": {
        "hostvars": {
            "centos1": {
                "ansible_host": "192.168.56.122",
                "ansible_user": "ealvarez"
            },
            "ubuntu1": {
                "ansible_host": "192.168.56.120"
            },
            "ubuntu2": {
                "ansible_host": "192.168.56.121"
            }
        }
    },
    "all": {
        "children": [
            "centos",
            "ubuntu",
            "ungrouped"
        ]
    },
    "centos": {
        "hosts": [
            "centos1"
        ]
    },
    "ubuntu": {
        "hosts": [
            "ubuntu1",
            "ubuntu2"
        ]
    }
}
```

```
alvarezworkstation@ubuntu1:~/ansible-lab$ ansible all --list-hosts
[WARNING]: No inventory was parsed, only implicit localhost is available
[WARNING]: provided hosts list is empty, only localhost is available. Note that
the implicit localhost does not match 'all'
  hosts (0):
```

- Successful ping results

```
alvarezworkstation@ubuntu1:~/ansible-lab$ ansible all -m ping -i inventory.yml
ubuntu1 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3"
    },
    "changed": false,
    "ping": "pong"
}
ubuntu2 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3"
    },
    "changed": false,
    "ping": "pong"
}
[DEPRECATION WARNING]: Distribution centos 9 on host centos1 should use
/usr/libexec/platform-python, but is using /usr/bin/python for backward
compatibility with prior Ansible releases. A future Ansible release will
default to using the discovered platform python for this host. See https://docs
.ansible.com/ansible/2.10/reference_appendices/interpreter_discovery.html for
more information. This feature will be removed in version 2.12. Deprecation
warnings can be disabled by setting deprecation_warnings=False in ansible.cfg.
centos1 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python"
    },
    "changed": false,
    "ping": "pong"
}
alvarezworkstation@ubuntu1:~/ansible-lab$ ansible all -m ping -i inventory.yml
ubuntu2 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3"
    },
    "changed": false,
    "ping": "pong"
}
ubuntu1 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3"
    },
    "changed": false,
    "ping": "pong"
}
```

Part C: Creating a Metagroup

1. Modify the inventory file.
2. Test the metagroup.
3. Display all hosts belonging to the metagroup.

Demonstrate successful execution of the metagroup.

```
alvarezworkstation@ubuntu1:~/ansible-lab$ ansible linuxservers --list-hosts -i i
nventory.yml
  hosts (3):
    ubuntu1
    ubuntu2
    centos1
alvarezworkstation@ubuntu1:~/ansible-lab$
```

Part D: Creating Variables

Demonstrate successful use of variables.

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

3. CONCLUSION

In this prelim exam, I understand how ssh works on each server. I just had a problem fixing my hostname in centos so I forced my yaml to read my centos's hostname to alvarez. That's why my inventory.yml is different compared to the procedure.

4. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

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Tools Used: **N/A**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **N/A**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

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